IB MATH AA SL

Lesson 4

Sum to Infinity & Applications

Number & Algebra · 60-minute lesson

Syllabus SL 1.8 ·

Do Now — recall

ON YOUR WHITEBOARD

1. Work out ½ + ¼ + ⅛ + 1/16
2. What value do the partial sums get close to?
3. In 8, 4, 2, 1, … what is r?

Recall:

$$S_n = \frac{u_1(1 - r^n)}{1 - r}$$

Answers (click to reveal)

1. 15/16
2. it approaches 1
3. r = ½

What we're learning today

LEARNING OBJECTIVES

1. **Understand**
why a geometric series can have a finite sum when $|r| < 1$.

2. **State**
the condition for convergence: $|r| < 1$.

3. **Apply**
$S_\infty = u_1/(1-r)$.

4. **Solve**
problems including recurring decimals and real contexts.

SUCCESS CRITERIA — I CAN...

- ✓ state when a geometric series converges
- ✓ use the sum-to-infinity formula
- ✓ convert a recurring decimal to a fraction
- ✓ rearrange S_∞ to find u_1 or r

Sum to infinity

THE FORMULA

$$S_\infty = \frac{u_1}{1-r}$$

Condition

only valid when $|r| < 1$

Why

$r^n \to 0$ as $n \to \infty$, so the $(1-r^n)$ term $\to 1$

DOES IT CONVERGE?

$|r| < 1$

yes — the sum settles to a value

$|r| \geq 1$

no — the sum grows without limit

Check r first

always test the ratio.

Find S_∞

QUESTION

Find the sum to infinity of

$8 + 4 + 2 + 1 + \ldots$

SOLUTION

Step 1 $u_1 = 8$, $r = \frac{1}{2}$ ($|r| < 1$ ✓)

Step 2 substitute:

$$S_\infty = \frac{8}{1 - \frac{1}{2}}$$

Step 3 evaluate:

$$S_\infty = \frac{8}{\frac{1}{2}} = 16$$

Tip:

Always check $|r| < 1$ before using the formula — state it for the reasoning mark.

Your turn — S_∞

ANSWER IN YOUR BOOK

1. Find S_∞ of $9 + 3 + 1 + \dots$
2. Find S_∞ of $20 + 10 + 5 + \dots$
3. Does $2 + 4 + 8 + \dots$ converge? Why?

Answers (click to reveal)

1. $9/(1-⅓)=13.5$
2. $20/(1-½)=40$
3. No — $r = 2$, $|r| \geq 1$

Recurring decimals

AS A GEOMETRIC SERIES

$$0.3 = \frac{3}{10} + \frac{3}{100} +$$

Each place is ÷10 of the last → r = 1/10

Then

use S∞ to get the fraction.

WORKED IDEA

$0.\bar{3}$: $u_1 = 3/10$, $r = 1/10$

$S_\infty = (3/10)/(1-1/10)$

$= (3/10)/(9/10) = 1/3$

Find u_1 from S_∞

QUESTION

A convergent geometric series has $r = 0.2$ and sum to infinity 25.

Find the first term u_1.

SOLUTION

Step 1 rearrange $S_\infty = u_1/(1-r)$:

$$u_1 = S_\infty (1-r)$$

Step 2 substitute $r = 0.2$, $S_\infty = 25$:

$$u_1 = 25(1-0.2) = 25(0.8)$$

Step 3 evaluate:

$$u_1 = 20$$

EXAM-STYLE QUESTION

Exam-style question

A geometric series has first term 12 and common ratio r, with sum to infinity 20.

(a) Show that $r = 0.4$.   [3]

(b) Find the sum of the first 4 terms.   [3]

MARK SCHEME — reveal after students attempt (tutor only)

(a) $20 = 12/(1-r)$ (M1) → $1-r = 0.6$ (A1) → $r = 0.4$ (AG)

(b) $S_4 = 12(1-0.4^4)/(1-0.4)$ (M1)(A1) = 19.49 (4 s.f.) (A1)

Common mistakes

× **Forgetting $|r|<1$**

Check convergence before using S_∞.

× **$1-r$ sign**

Denominator is $(1 - r)$, not $(r - 1)$.

× **Using S_n**

S_∞ has no n — don't put a power of r in it.

× **Decimal place**

For a recurring decimal, $r = 1/10$ (or $1/100$ for two digits).

Mixed practice

QUESTIONS

1. Find S_∞ of $16 + 8 + 4 + \ldots$
2. Find S_∞ of $6 - 3 + 1.5 - \ldots$
3. A series has $u_1 = 30$, $S_\infty = 50$. Find r.
4. Write $0.\bar{4}$ as a fraction.

Answers (click to reveal)

1. $16/(1-½)=32$
2. $6/(1+½)=4$
3. $50=30/(1-r) \rightarrow r=0.4$
4. $(4/10)/(9/10)=4/9$

Plenary & exit ticket

KEY FORMULAS

$$S_\infty = \frac{u_1}{1 - r}$$

$$|r| < 1$$

EXIT TICKET — before you leave

1. Find S_∞ of $10 + 2 + 0.4 + \ldots$
2. Does $3 + 6 + 12 + \ldots$ converge?
3. A series has $r = \frac{1}{2}$, $S_\infty = 24$. Find u_1.

Answers (click to reveal)

1. $10/(1-0.2)=12.5$
2. No — $r = 2$
3. $u_1 = 24(1-\frac{1}{2}) = 12$

Self-assess:

- need more help
- nearly there
- confident

Homework

SET ON DR FROST MATHS (auto-marked)

Task:
Sum to infinity.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 5

Exponents & laws of indices — the rules that power all of this.

$$a^m \times a^n = a^{m+n}$$